Dose calculator: model and numerical data

Technical description of `dose_calculator.py`

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1 Quantity calculated and inputs

The script estimates uncollided-primary-photon ambient dose equivalent $H^*(10)$ at the endpoint of one ray, in mrem per laser pulse. It accepts Al, simplified stainless steel (SS), or polytetrafluoroethylene (Teflon/PTFE) as the thick target. Configuration is done by editing `CONFIG` in the script, or by passing a complete replacement with `-config FILE.json`. The output echoes every configuration input, reports important calculated quantities, and then prints the labeled dose result.

Canonical SI is used for configuration geometry, time, angle, and internal photon transport. Electron and photon energies remain in keV because the empirical source law and both radiation tables are tabulated in keV.

Input	Unit / choices
`laser.pulse_energy_J`	J
`laser.pulse_duration_s`	s, intensity FWHM
`laser.wavelength_m`	m
`laser.beam_fwhm_diameter_m`	m, at lens
`laser.focal_length_m`	m
`laser.M_squared`	dimensionless, ≥ 1
`target`	`Al`, `SS`, or `Teflon`
`source.observation_angle_rad`	rad, $0 \ldots \pi/2$
`path[*].material`	`vacuum`, `air`, `Al`, `SS`, `BK7`, or `BaF2`
`path[*].thickness_m`	m, along the observation ray

Specify `path` as an ordered list of layers from the source to the observation point, with one `material` and one nonnegative `thickness_m` per layer. Every thickness is measured along the observation ray; their sum is the source–observer distance r, and vacuum contributes to r but not attenuation. Layer order does not affect this homogeneous, uncollided-beam model when the same materials, thicknesses, and total distance are retained.

Material densities and the source coupling, temperature override, and angular width are deliberately stored outside `CONFIG` in `DENSITY_KG_M3` and `SOURCE_MODEL`. They are advanced model constants, not routine inputs. Normal output additionally reports w_0, I_0, Q, T, $\langle E \rangle$, $g(\theta)$, and r before the dose.

2 Laser and hot-electron model

For pulse energy U_L, temporal intensity FWHM τ, lens-plane intensity-FWHM diameter D, focal length f, wavelength λ, and the numerical beam-quality value $\mathcal{M} = M^2$,

$$w_L = D/\sqrt{2\ln 2}, \qquad w_0 = \mathcal{M}\lambda f/(\pi w_L), \tag{1}$$

$$\tau_{\text{eff}} = \tau\sqrt{\pi}/(2\sqrt{\ln 2}), \qquad I_0 = U_L/[(\pi w_0^2/2)\tau_{\text{eff}}] \quad [\text{W/m}^2]. \tag{2}$$

The assumed intensity is $I(\varrho, t) = I_0 e^{-2\varrho^2/w_0^2} e^{-4\ln 2\, t^2/\tau^2}$. The legacy numerical parameter required by the empirical temperature law is

$$Q = 10^8 I_0[\text{W/m}^2]\lambda^2[\text{m}^2] = I_0[\text{W/cm}^2](\lambda[\mu\text{m}])^2. \tag{3}$$

With $u = Q/(1.37 \times 10^{18})$, the automatic temperature is [2]

$$T[\text{keV}] = \begin{cases} 6 \times 10^{-5} Q^{0.33}, & Q \leq 1.6 \times 10^{17}, \\ 511u/(\sqrt{1+u}+1), & Q > 1.6 \times 10^{17}. \end{cases} \tag{4}$$

The second expression equals $511(\sqrt{1+u}-1)$ but is numerically stable. The advanced `SOURCE_MODEL` constants are $\eta = 1.0$, temperature override `None` (use this scaling), and $\sigma = 0.7853981633974483$ rad. A non-null advanced temperature overrides the scaling.

With electron rest energy $m = 511$ keV, the normalized Maxwell–Jüttner kinetic-energy density is [4]

$$F(E;T) = \frac{(E+m)\sqrt{E(E+2m)}e^{-E/T}}{\int_0^\infty (E'+m)\sqrt{E'(E'+2m)}e^{-E'/T}\,dE'}, \tag{5}$$

$$\langle E\rangle = \int_0^\infty E F(E;T)\,dE. \tag{6}$$

For coupling fraction η and $c_E = 1.602176634\times10^{-16}$ J/keV,

$$U_e = \eta U_L, \qquad N_e = U_e/(c_E\langle E\rangle), \qquad dN_e/dE = N_e F(E;T). \tag{7}$$

3 Bremsstrahlung source

The target is assumed to stop every source electron. For an element of atomic number Z [3],

$$a_Z(E) = 6\times10^{-4} Z(E/1000), \qquad Y_Z(E) = a_Z/(1+a_Z), \tag{8}$$

where E is in keV. Thus $Y_{\mathrm{Al}} = Y_{13}$, while the explicit mixture surrogates are

$$Y_{\mathrm{SS}} = 0.738Y_{26} + 0.181Y_{24} + 0.081Y_{28}, \tag{9}$$

$$Y_{\mathrm{Teflon}} = 0.240183Y_6 + 0.759817Y_9. \tag{10}$$

The Teflon mass fractions correspond to carbon and fluorine in PTFE, as tabulated by NIST [9]. The same approximate mass-weighted elemental-yield rule used for SS is applied to Teflon. Teflon is a target option but not a `path[*].material` option. Target density is not used because the model assumes an electron-stopping target and omits self-absorption and electron transport. For a single electron,

$$B(k\mid E) = \begin{cases} 2Y(E)(1-k/E), & 0<k<E,\\ 0, & k\ge E,\end{cases} \qquad \int_0^E B\,dk = Y(E)E. \tag{11}$$

The photon-energy spectrum, integrated over the forward hemisphere, is

$$S(k) = \frac{2U_e}{\langle E\rangle}[A(k) - kC(k)] \quad [\mathrm{J/keV/pulse}], \tag{12}$$

$$A(k) = \int_k^\infty F(E)Y(E)\,dE, \qquad C(k) = \int_k^\infty F(E)Y(E)E^{-1}\,dE. \tag{13}$$

4 Angle, shielding, and dose

For advanced angular width σ and configured observation angle θ, both in radians,

$$g(\theta) = \frac{e^{-\theta^2/(2\sigma^2)}}{2\pi\int_0^{\pi/2} e^{-u^2/(2\sigma^2)}\sin u\,du}, \qquad \int_{\mathrm{forward}} g\,d\Omega = 1. \tag{14}$$

For layer density ρ_j in kg/m^3, ray length ℓ_j in m, and total mass attenuation coefficient $(\mu/\rho)_j$ in m^2/kg,

$$\chi(k) = \sum_{j\neq\mathrm{vacuum}} (\mu/\rho)_j(k)\rho_j\ell_j, qquad d\mathcal{T} = e^{-\chi}, \qquad r = \sum_j \ell_j. \tag{15}$$

Elemental coefficients are combined by the mass-additivity rule $(\mu/\rho)_{\mathrm{mix}} = \sum_i w_i(\mu/\rho)_i$. For oxide mass fractions x_o, stoichiometric counts n_{io}, and atomic weights A_i, the conversion used for BK7 is

$$w_i = \sum_o x_o \frac{n_{io}A_i}{\sum_j n_{jo}A_j}. \tag{16}$$

The exact mass fractions are

$$\mathbf{w}_{\mathrm{SS}} = \{\mathrm{Fe}: 0.738,\ \mathrm{Cr}: 0.181,\ \mathrm{Ni}: 0.081\}, \tag{17}$$

$$\mathbf{w}_{\mathrm{BaF_2}} = \{\mathrm{Ba}: 0.7832764008365862,\ \mathrm{F}: 0.2167235991634137\}, \tag{18}$$

$$\mathbf{w}_{\mathrm{BK7}} = \{\mathrm{Si}: 0.3231389993841853,\ \mathrm{O}: 0.48388281234709274,\ \mathrm{B}: 0.03338480543545398,\ \mathrm{Ba}: 0.02749656874894017, \tag{19}$$

$$\mathrm{Na}: 0.07715367482585571,\ \mathrm{K}: 0.052216516907371474,\ \mathrm{As}: 0.002726622351100653\}. \tag{20}$$

BaF_2 fractions follow from the formula and atomic weights Ba 137.327 and F 18.998403163. BK7 is a nominal BK7/K9 attenuation surrogate, not a claim about SCHOTT's proprietary N-BK7 recipe. Its published nominal oxide mass recipe is 69.13% SiO_2, 10.75% B_2O_3, 3.07% BaO, 10.40% Na_2O, 6.29% K_2O, and 0.36% As_2O_3 [6]; elemental conversion used atomic weights Si 28.085, O 15.999, B 10.81, Ba 137.327, Na 22.98976928, K 39.0983, and As 74.921595. `DENSITY_KG_M3` stores Al 2699, SS 7900, air 1.2048, BK7 2510, and BaF_2 4890 kg/m^3. The BK7 density is the SCHOTT N-BK7 value; the BaF_2 density is the Hellma value [7, 8].

The photon fluence and dose are

$$\frac{d\Phi}{dk} = \frac{g(\theta)S(k)e^{-\chi(k)}}{r^2 c_E k} \quad [\text{photons/m}^2/\text{keV/pulse}], \tag{21}$$

$$H^*_{\text{pulse}}(10)[\text{mrem/pulse}] = 10^5 \int_{10}^{20000} h(k)\frac{d\Phi}{dk}\,dk, \tag{22}$$

where h is in Sv m^2/photon and 1 Sv $= 10^5$ mrem. Embedded source values are pSv cm^2/photon and are multiplied by 10^{-16}. Embedded NIST values are cm^2/g and are multiplied by 0.1 to obtain m^2/kg. The reported integral excludes photons below 10 keV and above 20 MeV.

5 Numerics and scope

Positive tables use log–log interpolation,

$$y(k) = \exp\left[\ln y_i + \frac{\ln k - \ln k_i}{\ln k_{i+1} - \ln k_i}(\ln y_{i+1} - \ln y_i)\right], \tag{23}$$

with no extrapolation. Duplicate absorption-edge energies are shifted to $k(1 \mp 10^{-8})$ only in the interpolation grid. At the normal internal resolution multiplier 1, the electron grid has 90,000 logarithmic points from 10^{-5} keV to $\max(20000, 80T)$ keV; the photon grid has 18,000 logarithmic points from 10 to 20,000 keV plus every table knot and 18,000 keV. Trapezoidal quadrature evaluates all energy integrals. Angular quadrature uses zero absolute tolerance, 10^{-10} relative tolerance, and stops at the smaller of $\pi/(2\sigma)$ and 40 Gaussian widths. Temperatures outside 0.05–100,000 keV are rejected. The result is rejected if the 18–20 MeV interval contributes more than 10^{-5} of dose.

The model omits target self-absorption, finite-target electron escape, characteristic lines, scattered photons and buildup, charged-particle transport, cascades, photonuclear reactions, and neutron dose. It assumes a point source, an energy-independent angular law, and all source energy in the forward hemisphere. It is a screening model, not proof of shielding adequacy.

References

[1] J. H. Hubbell and S. M. Seltzer, NIST SRD 126, *X-Ray Mass Attenuation Coefficients*, `https://physics.nist.gov/PhysRefData/XrayMassCoef/cover.html`. The script uses μ/ρ, not μ_{en}/ρ.

[2] T. Liang et al., *Radiat. Prot. Dosim.* 172, 346–355 (2016), doi:10.1093/rpd/ncv505.

[3] J. Coderre, MIT OCW 22.55J, "Interactions of Light Charged Particles with Matter," p. 6, `https://ocw.mit.edu/courses/22-55j-principles-of-radiation-interactions-fall-2004/resources/energ_depos_elec/`.

[4] M. Mościbrodzka and C. F. Gammie, *ApJ* 968, 6 (2024), doi:10.3847/1538-4357/ad4636.

[5] A. Ferrari and M. Pelliccioni, photon $H^*(10)/\Phi$ table, `https://fismel.lnf.infn.it/wp-content/uploads/sites/14/2023/08/photonsa.pdf`.

[6] Nantong Haisheng Optical, nominal H-K9L (BK7-type) oxide composition, `https://www.haishengoptical.com/colorless_optical_glass-show-219.html`.

[7] SCHOTT N-BK7 data sheet (density 2.51 g/cm^3), `https://shop.schott.com/medias/schott-datasheet-n-bk7-eng.pdf`.

[8] Hellma Materials BaF_2 data sheet (density 4.89 g/cm^3), `https://www.hellma-materials.com/wp-content/uploads/2024/11/HM_Optische_Kristalle_BaF2_Data_Sheet_EN.pdf`.

[9] NIST STAR, composition of polytetrafluoroethylene (Teflon), `https://physics.nist.gov/cgi-bin/Star/compos.pl?matno=227`.

A Embedded attenuation coefficients: Al

These are original input rows, with photon energy converted from MeV to keV. Repeated nominal energies are the lower and upper sides of an absorption edge, in that order. Source: `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z13.html`.

k (keV)	μ/ρ (cm^2/g)
1	1185
1.5	402.2
1.5596	362.1
1.5596	3957
2	2263
3	788
4	360.5
5	193.4
6	115.3
8	50.33
10	26.23
15	7.955
20	3.441
30	1.128
40	0.5685
50	0.3681
60	0.2778
80	0.2018
100	0.1704
150	0.1378
200	0.1223
300	0.1042
400	0.09276
500	0.08445
600	0.07802
800	0.06841
1000	0.06146
1250	0.05496
1500	0.05006
2000	0.04324
3000	0.03541
4000	0.03106
5000	0.02836
6000	0.02655
8000	0.02437
10000	0.02318
15000	0.02195
20000	0.02168

B Embedded attenuation coefficients: Cr

These are original input rows, with photon energy converted from MeV to keV. Repeated nominal energies are the lower and upper sides of an absorption edge, in that order. Source: `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z24.html`.

k (keV)	μ/ρ (cm^2/g)
1	7405
1.5	2694
2	1277
3	433.9
4	198.8
5	108
5.9892	65.74
5.9892	597.7
6	516
8	251.3
10	138.6
15	45.71
20	20.38
30	6.434
40	2.856
50	1.55
60	0.9639
80	0.4905
100	0.3166
150	0.1788
200	0.1378
300	0.1067
400	0.09213
500	0.08281
600	0.07598
800	0.0662
1000	0.0593
1250	0.05295
1500	0.04832
2000	0.04213
3000	0.03559
4000	0.03235
5000	0.03057
6000	0.02956
8000	0.02869
10000	0.02855
15000	0.0292
20000	0.03026

C Embedded attenuation coefficients: Fe

These are original input rows, with photon energy converted from MeV to keV. Repeated nominal energies are the lower and upper sides of an absorption edge, in that order. Source: `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z26.html`.

k (keV)	μ/ρ (cm^2/g)
1	9085
1.5	3399
2	1626
3	557.6
4	256.7
5	139.8
6	84.84
7.112	53.19
7.112	407.6
8	305.6
10	170.6
15	57.08
20	25.68
30	8.176
40	3.629
50	1.958
60	1.205
80	0.5952
100	0.3717
150	0.1964
200	0.146
300	0.1099
400	0.094
500	0.08414
600	0.07704
800	0.06699
1000	0.05995
1250	0.0535
1500	0.04883
2000	0.04265
3000	0.03621
4000	0.03312
5000	0.03146
6000	0.03057
8000	0.02991
10000	0.02994
15000	0.03092
20000	0.03224

D Embedded attenuation coefficients: Ni

These are original input rows, with photon energy converted from MeV to keV. Repeated nominal energies are the lower and upper sides of an absorption edge, in that order. Source: `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z28.html`.

k (keV)	μ/ρ (cm^2/g)
1	9855
1.00404	9753
1.0081	9654
1.0081	10990
1.5	4234
2	2049
3	709.4
4	328.2
5	179.3
6	109
8	49.52
8.3328	44.28
8.3328	329.4
10	209
15	70.81
20	32.2
30	10.34
40	4.6
50	2.474
60	1.512
80	0.7306
100	0.444
150	0.2208
200	0.1582
300	0.1154
400	0.09765
500	0.08698
600	0.07944
800	0.06891
1000	0.0616
1250	0.05494
1500	0.05015
2000	0.04387
3000	0.03745
4000	0.03444
5000	0.03289
6000	0.0321
8000	0.03164
10000	0.03185
15000	0.0332
20000	0.03476

E Embedded attenuation coefficients: air

These are original input rows, with photon energy converted from MeV to keV. Repeated nominal energies are the lower and upper sides of an absorption edge, in that order. Source: `https://physics.nist.gov/PhysRefData/XrayMassCoef/ComTab/air.html`.

k (keV)	μ/ρ (cm^2/g)
1	3606
1.5	1191
2	527.9
3	162.5
3.2029	134
3.2029	148.5
4	77.88
5	40.27
6	23.41
8	9.921
10	5.12
15	1.614
20	0.7779
30	0.3538
40	0.2485
50	0.208
60	0.1875
80	0.1662
100	0.1541
150	0.1356
200	0.1233
300	0.1067
400	0.09549
500	0.08712
600	0.08055
800	0.07074
1000	0.06358
1250	0.05687
1500	0.05175
2000	0.04447
3000	0.03581
4000	0.03079
5000	0.02751
6000	0.02522
8000	0.02225
10000	0.02045
15000	0.0181
20000	0.01705

F Additional elemental attenuation coefficients for BK7 and BaF_2

Exact embedded NIST μ/ρ inputs over the calculated 10–20,000 keV range. Sources are the NIST element pages for B, O, F, Na, Si, and K.

k (keV)	B	O	F	Na	Si	K
10	1.255	5.952	8.205	15.57	33.89	79.07
15	0.4827	1.836	2.492	4.694	10.34	25.03
20	0.3014	0.8651	1.133	2.057	4.464	10.93
30	0.2063	0.3779	0.4487	0.7197	1.436	3.413
40	0.1793	0.2585	0.2828	0.3969	0.7012	1.541
50	0.1665	0.2132	0.2214	0.2804	0.4385	0.8679
60	0.1583	0.1907	0.1920	0.2268	0.3207	0.5678
80	0.1472	0.1678	0.1639	0.1796	0.2228	0.3251
100	0.1391	0.1551	0.1496	0.1585	0.1835	0.2345
150	0.1243	0.1361	0.1298	0.1335	0.1448	0.1582
200	0.1136	0.1237	0.1176	0.1199	0.1275	0.1319
300	0.09862	0.1070	0.1015	0.1029	0.1082	0.1080
400	0.08834	0.09566	0.09073	0.09185	0.09614	0.09495
500	0.08065	0.08729	0.08274	0.08372	0.08748	0.08600
600	0.07460	0.08070	0.07649	0.07736	0.08077	0.07922
800	0.06549	0.07087	0.06717	0.06788	0.07082	0.06929
1000	0.05890	0.06372	0.06037	0.06100	0.06361	0.06216
1250	0.05266	0.05697	0.05399	0.05454	0.05688	0.05556
1500	0.04791	0.05185	0.04915	0.04968	0.05183	0.05068
2000	0.04108	0.04459	0.04228	0.04282	0.04480	0.04399
3000	0.03284	0.03597	0.03422	0.03487	0.03678	0.03666
4000	0.02798	0.03100	0.02960	0.03037	0.03240	0.03282
5000	0.02476	0.02777	0.02663	0.02753	0.02967	0.03054
6000	0.02248	0.02552	0.02457	0.02559	0.02788	0.02915
8000	0.01945	0.02263	0.02195	0.02319	0.02574	0.02766
10000	0.01755	0.02089	0.02039	0.02181	0.02462	0.02704
15000	0.01495	0.01866	0.01846	0.02023	0.02352	0.02687
20000	0.01368	0.01770	0.01769	0.01970	0.02338	0.02737

As: μ/ρ (cm^2/g)		Ba: μ/ρ (cm^2/g)	
k (keV)	value	k (keV)	value
10	41.15	10	186.0
11.8667	25.77	15	63.47
11.8667	179.2	20	29.38
15	98.56	30	9.904
20	45.64	37.4406	5.498
30	15.06	37.4406	29.19
40	6.760	40	24.57
50	3.635	50	13.79
60	2.203	60	8.511
80	1.030	80	3.963
100	0.5971	100	2.196
150	0.2622	150	0.7828
200	0.1719	200	0.4046
300	0.1150	300	0.1891
400	0.09414	400	0.1265
500	0.08259	500	0.09923
600	0.07483	600	0.08410
800	0.06440	800	0.06744
1000	0.05735	1000	0.05803
1250	0.05106	1250	0.05058
1500	0.04661	1500	0.04592
2000	0.04093	2000	0.04078
3000	0.03539	3000	0.03692
4000	0.03296	4000	0.03598
5000	0.03187	5000	0.03612
6000	0.03141	6000	0.03669
8000	0.03146	8000	0.03844
10000	0.03207	10000	0.04042
15000	0.03405	15000	0.04518
20000	0.03603	20000	0.04902

The As and Ba source pages are `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z33.html` and `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z56.html`. Repeated energies are the lower and upper sides of an absorption edge.

G Embedded photon dose-response coefficients

Original selected $H^*(10)/\Phi$ values from Ref. [5], converted to keV in the first column. These are not H_{MAX}/Φ values.

k (keV)	$h(k)$ (pSv cm^2)
10	0.083
15	0.85
20	1.05
30	0.8
40	0.62
50	0.52
60	0.51
80	0.56
100	0.62
150	0.87
200	1.23
300	1.81
400	2.36
500	2.78
600	3.46
800	4.29
1000	5.18
1500	6.92
2000	8.25
3000	10.4
4000	10.7
5000	10.4
6000	9.58
8000	9.1
10000	8.76
20000	8.29